Total No. of Questions : 8]

SEAT No. :

PD-4488

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T.E. (Artificial Intelligence and Machine Learning) (Honors)
ARTIFICIAL INTELLIGENCE
(2019 Pattern) (Semester - VI) (310303)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7or Q8.***
2) ***Neat diagrams must be drawn whenever necessary.***
3) ***Figures to the right indicate full marks.***

Q1) a) Represent the following sentences into formulas in predicate logic. **[9]**

i) John likes all kinds of food.

ii) Apples are food.

iii) Chicken are food.

iv) Anything anyone eats and isn't killed by is food.

v) Bill eats peanuts and is still alive.

vi) Sue eats everything Bill eats.

b) Explain Unification algorithm with suitable example. **[8]**

OR

Q2) a) Explain various operators used in propositional logic for knowledge base building. **[9]**

b) Explain Bayesian inference using a suitable example. **[8]**

Q3) a) Explain how Support vector Machines are used for classification with suitable example. **[6]**

b) Explain how Decision Trees are used in Learning. **[6]**

c) Explain : **[6]**

i) Supervised Learning

ii) Unsupervised Learning

P.T.O.

OR

Q4) a) What is Artificial Neural Network? Give two applications of artificial neural networks in detail. **[6]**

b) With the help of an architecture diagram explain multilayer feed forward artificial neural network. **[6]**

c) Explain linear regression. Find linear regression equation for the following two sets of data : **[6]**

X	Y
2	3
4	7
6	5
8	10

Q5) a) Illustrate Mini-Max search for the tic-tac-toe game. **[9]**

b) Explain Alpha-Beta Pruning with an example. **[8]**

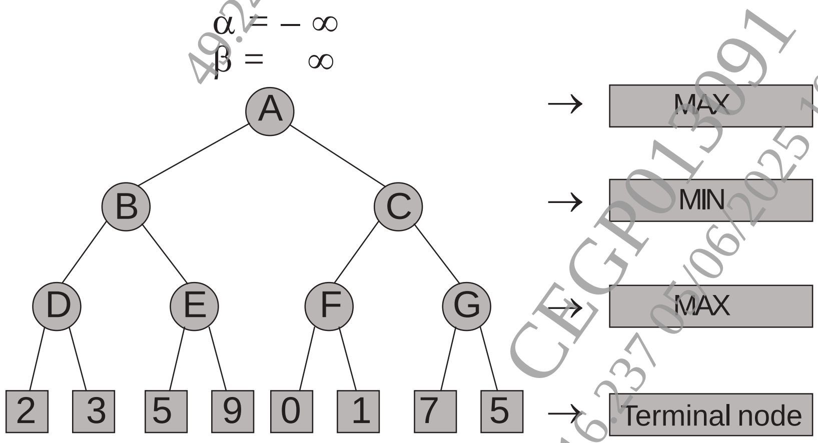
OR

Q6) a) Write anote on : **[9]**

i) Types of Games in AI.

ii) State-of-the-art Game Programs

b) Solve given two player search tree using Alpha-beta pruning. **[8]**



Q7) a) Explain forward chaining and backward chaining for a simple example. **[9]**

b) What is NLP? Explain all five phases of NLP. **[9]**

OR

Q8) a) Explain the applications of Natural Language Processing. **[9]**

b) Represent the architecture of an expert system. label the various components in the diagram and explain. **[9]**